Cellulose
https://doi.org/10.1007/s10570-019-02753-x

ORIGINAL RESEARCH

Production of cellulose aerogels from coir fibers via an alkali–urea method for sorption applications

Mar'atul Fauziyah · Widiyastuti Widiyastuti · Ratna Balgis · Heru Setyawan

Received: 10 March 2019 / Accepted: 9 September 2019

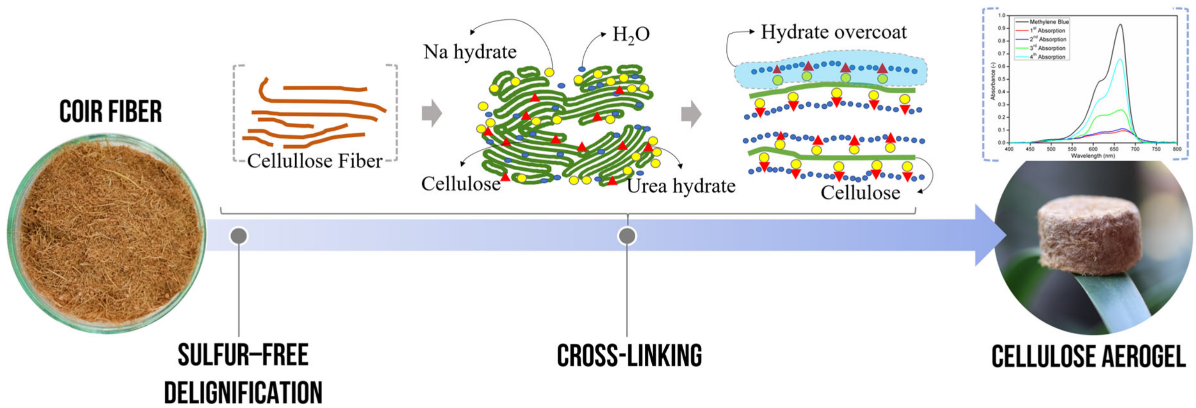
Abstract Biodegradable cellulose aerogels have been successfully prepared from coir fibers using a sulfur-free method and NaOH–urea system. Sulfur was avoided during pretreatment because it is environmentally harmful. Interestingly, these pretreatments had a strong effect on the physical properties of the aerogels produced. Good physical properties of the cellulose aerogels were obtained when the Kappa number, i.e., the lignin content, in the pulp was lower than 14.8. NaOH–urea played an important role in transforming cellulose I to cellulose II and crosslinked cellulose to form an aerogel structure. The aerogel had a macroporous structure, ultralight density, high porosity, good durability, and thermal stability. The aerogel was capable of absorbing 22 and 18 times its dry weight in water and oil, respectively. The material also had a high capacity for methylene blue dye adsorption of up to 62 g/g, which was one hundred times higher than that of adsorbents synthesized from the other natural matters. Therefore, the prepared aerogels have potential for various sorption applications.

M. Fauziyah · W. Widiyastuti · H. Setyawan (✉)
Department of Chemical Engineering, Faculty of Industrial Technology, Sepuluh Nopember Institute of Technology, Kampus ITS Sukolilo, Surabaya 60111, Indonesia
e-mail: sheru@chem-eng.its.ac.id
URL: https://elkimkor.com

R. Balgis
Department of Chemical Engineering, Graduate School of Engineering, Hiroshima University, 1-4-1 Kagamiyama, Higashi-Hiroshima, Hiroshima 739-8527, Japan

Published online: 14 September 2019

Graphic abstract



Keywords NaOH–urea system · Absorbent · Adsorbent · Coir fibers · Cellulose aerogel

Introduction

Aerogels are highly porous materials with distinct properties, such as ultralight weight, small pore size (~ 1–50 nm), high surface area, and strong adsorption (Demilecamps et al. 2014; Shen et al. 2017). Owing to their intriguing properties, aerogels have very wide potential applications ranging from scaffolds and matrices for controlled drug release to gas absorbers, flame retardants and thermal insulating materials, as well as carbons for batteries and fuel cell electrodes (Nguyen et al. 2013, 2014; Nazriati et al. 2014; Han et al. 2015; Liang et al. 2015; Li et al. 2017). Cellulose, being the most abundant biopolymer in nature and a major constituent of plant structures, has attracted much attention for making such aerogel materials. Coir fibers, an abundantly available form of agricultural waste, are rich in cellulose, with a content of approximately 23–43%, 3–12% of which is hemicellulose, embedded in a matrix of noncellulosic polysaccharides and lignin that accounts for approximately 35–54% (Maher et al. 2008). This material is a potential candidate for conversion into valuable cellulose aerogels. However, the methods to purify natural-origin cellulose materials to meet the requirements of feedstocks for first-class aerogel materials (high-purity cellulose I) typically require high temperature and pressure, consume large amounts of chemicals and cause significant loss of the two main cellulose components—lignin and hemicellulose (Lu et al. 2012).

Cellulose aerogels are typically prepared through a dissolution–coagulation route followed by drying, either supercritical or freeze drying (Gavillon and Budtova 2008; Aaltonen and Jauhiainen 2009; Olsson et al. 2010; Korhonen et al. 2011). However, when attempting to utilize most of the cellulose components in natural-origin cellulose materials for high-performance aerogels, there are some difficulties in dissolving the cellulose in the working medium. A solvent with high solubility for cellulose and lignin should be used. Ionic liquids (ILs) seem to meet this criterion (Lu et al. 2012; Chen et al. 2014; Shen et al. 2017; Mussana et al. 2018). However, IL-based cellulose systems are very expensive and have a high viscosity, which may limit their commercialization. Recently, dimethyl sulfoxide (DMSO) combined with LiCl (Chen et al. 2016; Liu et al. 2019), tetrabutylammonium fluoride (TBAF), and *N*-methylimidazole (NMI) (Lu and Ralph 2003) have also been used as cellulose solvents. Nevertheless, DMSO contains sulfur, which must be avoided for sorption applications due to environmental harm. Alternatively, a cheaper NaOH–urea aqueous system has been successfully used as a solvent to dissolve cellulose materials such as paper waste, cotton stalks and waste from cotton fabrics (Cai and Zhang 2006; Qi et al. 2013; Nguyen et al. 2014). This solution system has been used successfully to produce aerogels by coagulation and appropriate drying methods.

In the present work, a modified NaOH–urea method was used to prepare cellulose aerogels from coir fibers. The much higher content of lignin in coir fibers than in cotton stalks ($\sim$ 18%) (Budde et al. 2019), paper waste and waste from cotton fabrics prevents the NaOH–urea process from being adopted directly. The high lignin content might cause the aerogel produced to be stiff, brittle and difficult to squeeze. Therefore, a combination of mechanical and sulfur-free alkali treatment at atmospheric conditions was introduced as a preliminary step to remove some lignin from the coir fibers, resulting in a cellulose pulp suitable for use as a precursor for aerogels. The Kraft process, which is commonly used in modern pulp and paper production, was avoided here to reduce the lignin loss and to eliminate the presence of sulfur, which could hinder the application of the prepared aerogels as sorbents. The effect of different ratios of NaOH to urea on the physical properties of the cellulose aerogels was investigated. Simultaneously, a derived multifunctional aerogel with superior sorption properties for water, oil and dye was developed. Our proposed method is promising for preparing cellulose aerogels via a low-cost and facile production method because of its simple process and the use of coir fibers, which are sustainable and readily available at low cost.

Materials and methods

Materials

Coir fibers were collected from a coconut milling facility in Keputih, Surabaya, Indonesia. Sodium hydroxide (NaOH) and ethanol (C_2H_5OH) were supplied by Merck. Urea (CON_2H_4) was supplied by PT. Petrokimia Gresik, Indonesia. Demineralized water was used for all synthesis and treatment processes. All chemicals were reagent grade and were used as received without further purification.

Preparation of cellulose aerogels

A combination of mechanical and chemical treatments was used to remove part of the lignin from the coir fibers. The coir fibers were ground in a milling machine and sieved to 100–150 mesh. The ground coir fibers were then digested in alkali solution at boiling point under atmospheric reflux. The digestion was conducted in 6% NaOH solution with a volume ratio of 15–25 mL per gram of coir fibers for 4 h. The obtained pulp was washed with demineralized water to remove the sodium–lignin byproduct; thereafter, the pulp could be used as a precursor in preparing cellulose aerogels.

The coir fiber pulp was dispersed in an aqueous NaOH–urea solution by sonicating for 30 min. The NaOH–urea solution consisted of 1 g NaOH and 0–5 g urea dissolved in 10 mL of demineralized water. The mixture was placed in a refrigerator for 24 h to allow gelation. After the gel had set, it was thawed at room temperature and immersed in ethanol (98%) for coagulation. A glass beaker was used as a mold to obtain a specimen approximately 1 cm thick and 3.5 cm in diameter. After coagulation, the gel was immersed in demineralized water to exchange ethanol with water. The demineralized water was changed periodically each hour until the solution was free from ethanol. This was indicated by a change in the color of the water from brownish–yellow to colorless and no smell of ethanol. The sample was then frozen at $-$ 20 °C for 12 h prior to freeze drying. Freeze–drying of the frozen sample was carried out for 24 h.

Characterization

The content of α-cellulose and Kappa number of the pulp were determined using the Indonesian National Standards SNI 0444:2009 and SNI 0494:2008, respectively. The samples were measured in duplicate. Crosslinking of the cellulose aerogels by NaOH–urea was observed using Fourier transform infrared spectroscopy (FTIR; Shimadzu IRTracer-100) over the wavenumber range of 400–4000 cm^{-1}. The crystallinity of the material during treatment and synthesis was characterized using X-ray diffraction (XRD; PANalytical, X'Pert Pro) analysis performed over an angle range of 10°–40° with a step size of 0.05°. Different preparations were required for wet and dry samples prior to performing the XRD analysis. For the wet sample, the sample was spread on a flat glass surface and dried in an oven at a temperature of 80 °C. The dried sample was then peeled off, and the sample stuck on the glass surface was used for analysis. For the dry sample, the sample was soaked in liquid nitrogen and cut into a thin slab. The physical transformation of the coir fiber into cellulose aerogel was studied using color value, carbon content and viscosity changes. The color value was measured using the ICUMSA method by dissolving 0.2 g of sample in 100 mL of water with pH adjusted to 7.0 (Liu et al. 2016; Shen and Li 2016). The solution was filtered, and the absorbance of the filtrate at 420 nm was analyzed by UV spectrophotometer (Precise Perfect Model N4S). The color value was then calculated using the following equation:

$$\text{Color value} = \frac{A}{b \times c} \times 1000 \quad (1)$$

where A is the absorbance value at 420 nm, while b and c are the cuvette thickness (cm) and solution concentration (g/mL), respectively. The change in carbon content was measured by proximate analysis. The viscosity of the specimen during synthesis was measured using a B-shaped rotary viscometer. The morphology of the aerogel was observed by scanning electron microscopy (SEM; FEI Inspect S-50).

The bulk density of the prepared cellulose aerogel was calculated using the following equation:

$$\rho = \frac{m}{V} \quad (2)$$

where m and V are the mass (g) and volume (cm^3) of the aerogel, respectively. The porosity of the cellulose aerogel was calculated using the following equation:

$$P = \frac{V - m/\rho_c}{V} \quad (3)$$

where ρ_c is the density of bulk cellulose (1.528 g/cm^3) (Wang et al. 2015). The porosities of the samples were measured in triplicate. The specific surface area and pore volume of the cellulose aerogel were quantitatively determined using the N_2 adsorption–desorption method (NOVA 1200, Quantachrome). The aerogel was degassed at 60 °C under flowing nitrogen for 5 h prior to the measurement. The specific surface area was determined by the Brunauer–Emmet–Teller (BET) method at a relative pressure < 0.3, and the total pore volume was calculated from the adsorption–desorption profiles.

The mechanical strength of the aerogel was evaluated using a compression test. This test was performed by measuring the change in height of the cellulose aerogel with variations in the compressed load. The change in height indicates the flexibility of the fiber bonds in the cellulose aerogel. The sample was placed on a flat base plate and first loaded with a 50 g weight, and the change in height was measured. This procedure was repeated by increasing the load in increments of 50 g until the loaded weight reached approximately 1000 g.

The thermal stability of the aerogel was analyzed by thermogravimetric analysis (TGA; Shimadzu Corp., 50/50H). The analysis was carried out from room temperature to 1000 °C at a rate of 5 °C/min in nitrogen atmosphere.

Water and oil absorption tests

The water absorption capacity of cellulose aerogel was determined by a water absorption test using a modified version of ASTM D570-98. The dimensions and weight of each sample were measured before and after each test. The dry sample was weighed and immersed in 800 mL of demineralized water for 30 min. After immersion, the wet sample was drawn up at a rate of 20 cm/min using a controlled motor. Excess water on the surface of the sample was removed with filter paper. The wet sample was weighed, dried or squeezed, and weighed again. The test was repeated three times.

The oil absorption capacity of the aerogel was determined by an oil absorption test using a modified version of ASTM F726-06. Oil with a viscosity of SAE 40 (API Service SE/CC) was used, and the test was carried out following the steps similar to those used for the water absorption test. Here, the excess oil was allowed to drain for 8 min after lifting the wet sample. Both water and oil absorption tests were performed in triplicate.

Dye adsorption experiment

Continuous adsorption experiments were performed using a monolith cellulose aerogel with a porosity of 0.969 and a diameter of 3 cm. The dye used in the adsorption experiments was methylene blue (MB; Merck, p.a.), and it was used without further purification. The aerogel height was increased from 0.92 to 2.85 cm to examine the effect of adsorbent height on the MB adsorption efficiency. MB (50 mg/L) dissolved in demineralized water was continuously fed to the monolith aerogel from the top at a flow rate of 5 mL/min using a syringe pump. All experiments were performed at ambient temperature. Samples were collected periodically at the bottom of the monolith aerogel, and the concentration was measured by a UV–Vis spectrophotometer (GENESYS 10S, Thermo Scientific) at a wavelength of 665 nm.

Results and discussion

Properties of cellulose aerogels

Because of its rigid structure and insolubility in water, the presence of lignin in aerogel can reduce the absorption performance when used as an absorbent (Laurichesse and Avérous 2014). The rigid structure makes the aerogel harder and inelastic, inhibiting squeezing to remove the absorbed liquid. In addition, lignin is a hydrophobic compound with low affinity for water that can inhibit the absorption of water by cellulose. Therefore, some parts of lignin, especially those with rigid structures and hydrophobic properties, must be removed from the coir fibers. For this purpose, both physical and chemical methods were used.

Figure 1 shows the effects of particle size on the α-cellulose content and Kappa number of the pulp at various volume to weight ratios of NaOH solution to coir fibers. As expected, the higher the volume to weight ratio of NaOH solution to coir fibers is, the lower the Kappa number of the pulp is. This may be caused by the capability of NaOH, which forms an alkali solution, to dissolve lignin (Grishechko et al. 2013; Salas et al. 2014). It is apparent that the size of the coir fibers also has a significant effect on the Kappa number. However, the effect is contradictory to that of the volume to weight ratio of solvent and solid. In this case, smaller fibers tend to decrease the Kappa number due to the higher active surface area, which enables NaOH to bind lignin. However, the effect of the volume to weight ratio of solvent to coir fibers on the α-cellulose content is different from that of coir fiber size. Hence, the objective of the treatment was to obtain the highest possible α-cellulose content and the lowest possible Kappa number. We found that the content of α-cellulose changed based on the treatment; the highest α-cellulose content (i.e., 92.5 wt%) was obtained at 120 mesh and a volume to weight ratio of solvent to solid of 20 mL/g. Under these conditions, the Kappa number of the pulp was approximately 14.8, which was a significant reduction (p value < 0.05) from that of the raw fiber (95.8). The Kappa number, as an indicator of the lignin contained in pulp, can be converted into lignin content using US Patent 5194388 (Foster and Forge 1993). The conversion results showed that this method could decrease the lignin content of the coir fiber from 14.08 to 2.18% for the pulp.

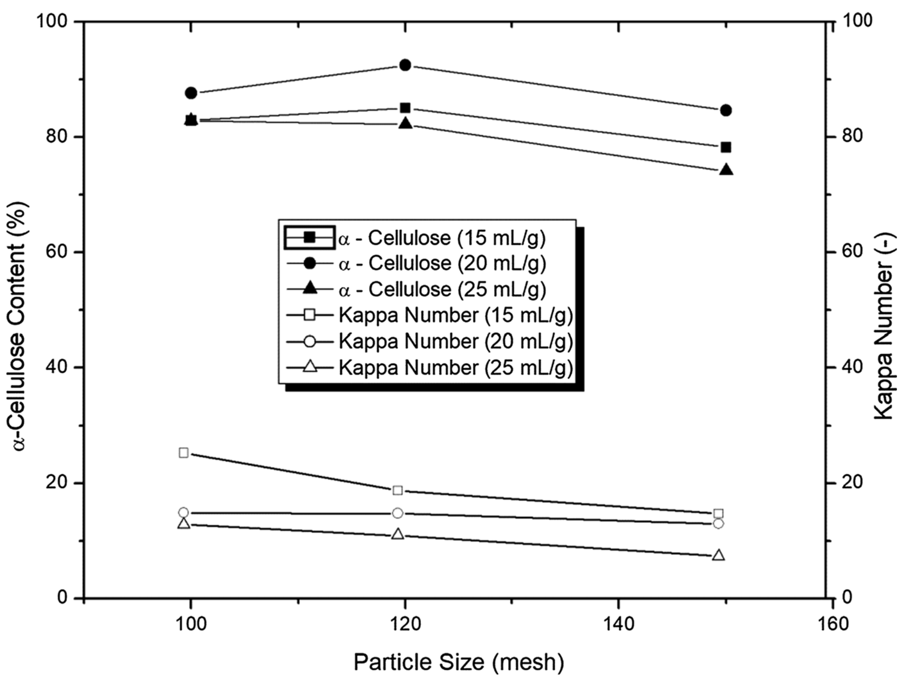


Fig. 1 Effect of particle size on the Kappa number and α-cellulose content of pulp at various volume to weight ratios of solvent to solid

It is worth noting that the NaOH–urea system can well dissolve cellulose materials in cotton stalks, waste from cotton fabric and paper waste, which have low lignin contents, and the aerogels produced have excellent properties. However, as discussed above, the lignin content in coir fibers is much higher. Although cellulose gels could also be produced at high lignin content, they were stiff, and the porosities were very low. In addition, they tended to be broken when squeezed. Aerogels with excellent properties were obtained when the Kappa number of the coir fiber pulp was at a maximum of 14.8. Therefore, the cellulose aerogels discussed in the following were prepared using coir fiber pulp with a Kappa number of approximately 14.8.

Figure 2 shows the FTIR spectra of three different samples: (a) pulp obtained after delignification; (b) pulp after adding NaOH–urea; and (c) cellulose aerogel. For all samples, four main bands corresponding to cellulose can be observed. The band at 3525 cm^{-1} can be attributed to intramolecular hydrogen bonding (O–H stretching) of α-cellulose, and the band at 2943 cm^{-1} corresponds to C–H stretching in CH and CH_2 of cellulose compounds (Reddy et al. 2014; Kathirselvam et al. 2019). Similarly, the bands at 1637 cm^{-1} and 1433 cm^{-1} are attributed to C=O carbonyl stretching of acetyl groups in hemicellulose or α-keto carboxylic acid in lignin (Kathirselvam et al. 2019). However, the bands in the samples after NaOH–urea treatment and in the aerogel are shifted slightly to the right. These results indicate that the alkali hydrate, urea hydrate, and free water destroyed the intra- and intermolecular hydrogen bonding of cellulose and physically caused cellulose swelling in solution (Cai and Zhang 2005). In addition, these phenomena imply that NaOH–urea treatment chemically changes the structure of cellulose from cellulose I to cellulose II (Oh et al. 2005). A new band at 1660 cm^{-1} appears in the spectra of solid samples after the addition of NaOH and urea (Fig. 1b). The band can be attributed to the functional group of urea (RHN–CO–NHR) (Baldanza et al. 2018). This urea band also appears in the spectrum of the cellulose aerogel (Fig. 1c) but is slightly shifted to the right, which may be caused by the dissolution of urea with cellulose. Thus, it appears that the addition of NaOH–urea plays an important role in the formation of the aerogel structure, in which NaOH–urea changes the structure of cellulose physically and chemically.

To obtain better insight into the transformation of cellulose structure during aerogel formation, XRD patterns of samples after each treatment were

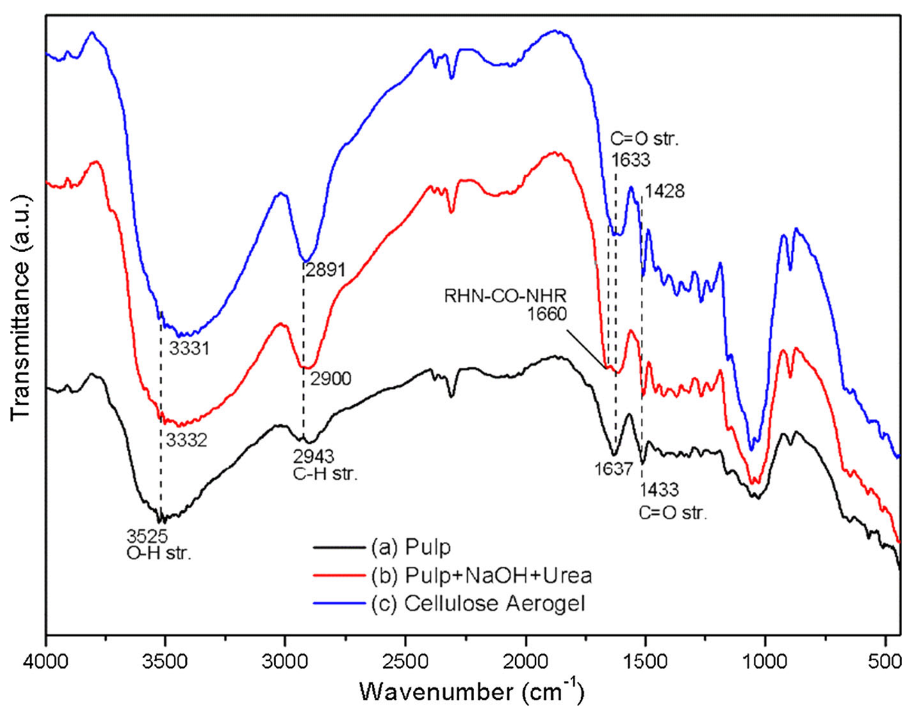


Fig. 2 Fourier transform infrared spectroscopy (FTIR) spectra of **a** pulp; **b** pulp + NaOH + urea; and **c** cellulose aerogel

examined. Figure 3 shows the XRD patterns of (a) pulp, (b) pulp treated with NaOH and urea, and (c) cellulose aerogel. The XRD pattern for coir pulp (Fig. 3a) has three characteristic peaks at 16.08°, 22.23°, and 34.3°, which match the pattern of cellulose I (Cai and Zhang 2005; French 2014; Wan et al. 2019). The broad peak at 16.08° was the overlapping of $(1\bar{1}0)$ and (110) planes (French 2014; Li et al. 2019). The pattern of pulp after NaOH–urea treatment (Fig. 3b) slightly changes becoming a little sharper and shifting to the left, to 15.28° and 21.88°. The pattern could be indicated as Na–cellulose. NaOH solution with the concentration ranging from 1.8–16.3 N took a considerable contribution to the conversion of cellulose into Na–cellulose (Okano and Sarko 1984; Sarko et al. 1987; Wertz et al. 2010). In this work, 2.5 N NaOH was used along with urea and ultrasonication treatment, hence could also convert the cellulose into Na–cellulose. The resulted pattern (Fig. 3b) also has a similar tendency with the pattern of Na–carboxymethyl cellulose (Socha et al. 2013). In the case of cellulose aerogel, the (110) plane clearly appears at 20.28°, while the position of the (020) plane appears at 21.38°. There is also a weak peak at 12.73° which can be assigned as $(1\bar{1}0)$ plane. These three peaks are the characteristic peaks of cellulose II (Cai and Zhang 2005; French 2014; Wan and Li 2016; Yu et al. 2017). Dissolution of cellulose into NaOH–urea at low temperature followed by regeneration could change the structure of cellulose I into cellulose II (Chen et al. 2015; Wan et al. 2019). All the diffraction peaks are broad, indicating the amorphous phase of cellulose,

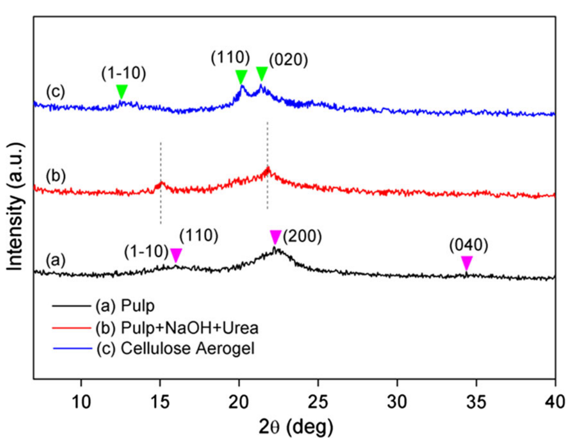


Fig. 3 X-ray diffraction (XRD) patterns of **a** pulp; **b** pulp + NaOH + urea; and **c** cellulose aerogel

which may be caused by the reduced periodic order within crystals (paracrystallinity) or the reduced crystallite size (French and Santiago Cintrón 2013; Ioelovich et al. 2010). However, following French and Santiago Cintrón (2013) and by the fact that the crystallite size of cellulose pulp, Na–cellullose and cellulose aerogel is very small (5.9, 2.1, and 7.2 nm, respectively), it is apparent that the crystallite size is the major factor for the broad peaks. For the quantitative approach, the crystallinity index was also calculated based on the XRD patterns using the Segal equation which was commonly used for cellulose I and also extended for cellulose II (Segal et al. 1959; Thygesen et al. 2005; Nam et al. 2016)

$$x_{CR} = \frac{I_t - I_{AM}}{I_t} \times 100 \quad (4)$$

where x_{CR} represents the crystallinity of the sample, I_t is the total intensity of the (200) peak for cellulose I and of the (020) peak for cellulose II, I_{AM} is the minimum between the (110) and (200) peaks at around 18° for cellulose I, and between the $(1\bar{1}0)$ and (110) peaks at around 16° for cellulose II (Azubuike et al. 2012; Nam et al. 2016). The crystallinity index changed significantly. The pulp (Fig. 3a) had a crystallinity index of 49.83%, which decreased after NaOH–urea treatment (Fig. 3b) to 34.01%. This might be caused by the crosslinking of celluloses (Yu et al. 2017). Urea has the ability to reduce the crystallinity of cellulose through van der Waals forces and to improve its solubility by increasing the fraction that is soluble in aqueous alkali solution by 1.5–2.5 times (Isobe et al. 2013; Xiong et al. 2014). Moreover, the XRD pattern of cellulose aerogel (Fig. 3c) became sharper, with a crystallinity index of 79.02%, which was the highest among the samples. The higher crystallinity index confirmed that cellulose I was transformed into cellulose II (French and Santiago Cintrón 2013). This result might indicate that the cellulose I of the pulp had been converted into cellulose II of cellulose aerogel through some Na–cellulose complexes pathway. Therefore, it can be concluded from the FTIR spectra and corroborated by the XRD patterns as discussed above that the obtained coir fiber pulp is cellulose I and it changes to cellulose II after NaOH–urea treatment.

Cellulose II is also known as the cellulose that passes through the mercerization process, which involves treatment with concentrated alkaline solution

(∼ 20%) followed by washing in water. Mercerization of cellulose is accompanied by fiber swelling, causing the cellulose suspension to be more viscous (Wertz et al. 2010). Therefore, the viscosity of a cellulose suspension can indicate cellulose transformation. The viscosity of the coir fiber suspension before delignification was initially approximately 625 cSt. After delignification, the obtained pulp was dispersed in aqueous NaOH–urea solution, resulting in a twofold increase in viscosity to 1173 cSt. The increase in viscosity might be caused by the swelling of fibers during gelation. This result confirmed the FTIR and XRD results, which indicated that cellulose I was changed into cellulose II by the mercerization process.

It is widely accepted that there is no strong direct interaction between urea and cellulose, and urea does not significantly affect the structural dynamics of water, except in aqueous alkali solutions (Isobe et al. 2013). Therefore, in the present study, cellulose–urea crosslinking was initiated by NaOH. NaOH–urea aqueous solution forms alkali hydrate, urea hydrate, and free water in a cellulose dispersion. The hydroxyl group (OH^-) of NaOH breaks the inter- and intramolecular hydrogen bonds of cellulose, as demonstrated by the FTIR spectra, and sodium (Na^+) hydration stabilizes the hydrophilic hydroxyl group. Subsequently, the urea hydrate can interact with the cellulose. Urea prevents the dissolved cellulose molecules from reaggregating by accumulating in the hydrophobic region, and the alkali hydrate penetrates the crystalline region (Isobe et al. 2013). Then, the cellulose chain takes up the alkali and urea hydrates, forming a swollen gel coated by the hydrates (Cai and Zhang 2005). After freeze drying, the hydrates sublimate the remaining cellulose, which forms crosslinking and yields a three-dimensional interconnected network called an aerogel structure.

To obtain better insight into the transformation of coir fibers to cellulose aerogel, the changes in lignin and carbon contents and color value were measured for the coir fibers, pulp and cellulose aerogel. Figure 4 shows the lignin and carbon contents and color values of the coir fibers, pulp, and cellulose aerogel. As mentioned above, chemical delignification successfully reduced the lignin content from 14.07% in the coir fibers to 2.18% in the pulp. Surprisingly, the NaOH–urea treatment further decreased the lignin content to 1.02%. It seems that NaOH can dissolve

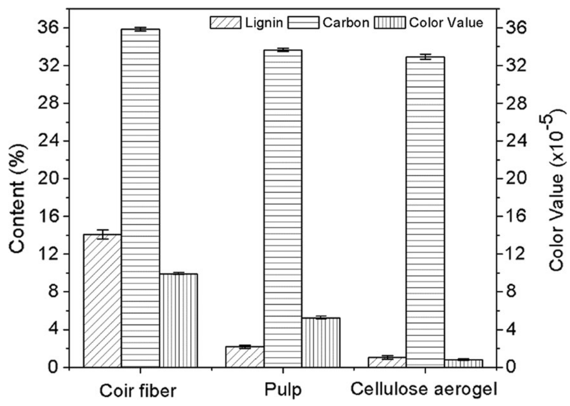


Fig. 4 Chemical content and color value of coir fiber, pulp, and cellulose aerogel

some lignin, which is removed during solvent exchange. Along with the decrease in lignin content, the carbon content also decreased from 35.87% in coir fibers to 33.65% in pulp and eventually to 32.92% in cellulose aerogel. Lignin contains a large amount of carbon, which also decreased following the decrease in lignin content. Moreover, the decrease in color value followed the trend of lignin and carbon contents. The color value decreased from 9.9×10^5 for coir fibers to 5.25×10^5 after delignification and finally to 0.8×10^5 for the cellulose aerogel. A decrease in color value means that the color becomes bright, which indicates that the content of lignin (dark brown) is significantly reduced. The cellulose aerogel is light brown with a smooth surface and highly porous structure. Although a small amount of lignin remains in the cellulose aerogel, the performance of the cellulose aerogel as an adsorbent is still excellent, as will be discussed later.

Figure 5 shows the SEM images of the prepared cellulose aerogel. The aerogel has a highly porous structure consisting of a network of interconnected uniform cellulose fibers, as depicted in Fig. 5a. The structure comprises a three-dimensional network of fibers forming macropores, each with a diameter of 20–200 µm. This network structure, which was formed by the cross-linking of cellulose as mentioned above, confirmed that NaOH–urea has a good dissolution capacity. Figure 5b–c shows longitudinal and cross sectional images of the cellulose fiber. The fibers themselves have a cylinder-like structure with a diameter of 40–70 µm and are successfully self-assembled via hydrogen bonding to form a network

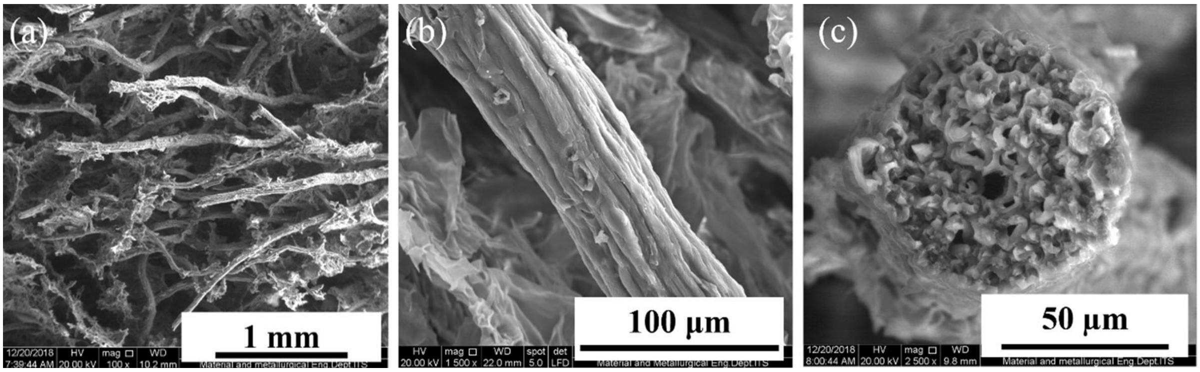


Fig. 5 SEM images of **a** cross-sectional area of cellulose aerogel; **b** longitudinal section of cellulose fiber; and **c** cross-sectional area of cellulose fiber

of open pores. The fibers also have pores with diameters of 2–4 µm, as depicted in Fig. 5c. These internal fiber pores also contributed to increasing the porosity of the aerogels.

Figure 6a presents the effect of the NaOH:urea ratio on the density and porosity of the cellulose aerogels. Urea had a significant effect (p value < 0.05) on the density and porosity of the aerogels, in which more porous matrices were formed when a higher concentration of urea was added. This arises because urea enhances the capability of NaOH to dissolve cellulose and increases the stability of the cellulose (Isobe et al. 2013). NaOH and urea affected not only the chemical process during synthesis but also the physical characteristics of the cellulose aerogel. They chemically destroyed the hydrogen bonds in the cellulose and crosslinked the components to produce a porous matrix, forming an aerogel (Sescousse et al. 2011; Isobe et al. 2013; Xiong et al. 2014). The prepared cellulose aerogels have ultralow density (approximately 0.047–0.091 g/cm^3) and high porosity (0.940–0.969), which are comparable to those of cellulose aerogels prepared from crude bagasse (0.088 g/cm^3 and 0.942, respectively) (Chen et al. 2016), nanocellulose (0.020 g/cm^3 and 0.980) (Pääkkö et al. 2008), waste paper (0.040 g/cm^3 and 0.973) (Nguyen et al. 2013), and bamboo fiber (0.054 g/cm^3 and 0.970) (Wan et al. 2015a). Moreover, BET analysis (Fig. 6b) revealed that the obtained cellulose aerogel has a specific surface area of 69.92 m^2/g with a pore volume of 0.078 cm^3/g, which is higher than that of cellulose aerogels prepared from nanocellulose (20 m^2/g) (Pääkkö et al. 2008) and coconut shell (9.1 m^2/g and 0.025 cm^3/g,

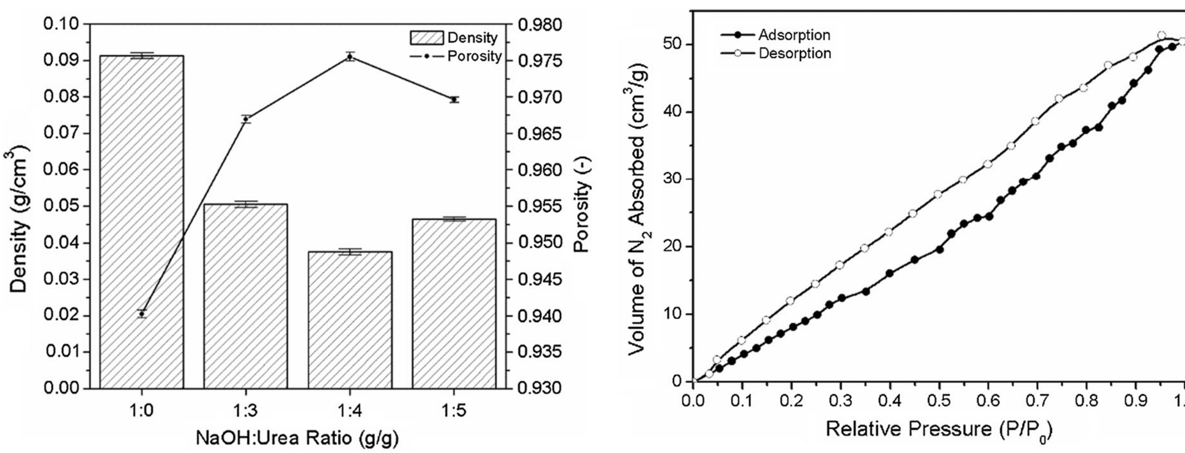


Fig. 6 **a** Effect of the NaOH:urea ratio on the density and porosity of the cellulose aerogels and **b** adsorption–desorption isotherm of cellulose aerogel

respectively) (Wan et al. 2015b) and comparable to that of lignocellulose aerogel prepared from lignocellulose/LiCl/DMSO solution (69.14 m^2/g) (Liu et al. 2019). Therefore, these physical properties confirmed that the cellulose aerogels prepared from coir fiber have the essential characteristics of a good absorbent.

Figure 7 shows the stress–strain curves of the aerogels prepared from the coir fibers. The strain increased proportionally to the applied stress. Young's modulus was calculated from the slope of the stress–strain curve in the elastic zone (the first linear zone), as presented in the inset of Fig. 7. The addition of urea reduced the Young's modulus of the aerogel and enabled NaOH to dissolve the cellulose, which became amorphous as mentioned above in the XRD and crosslinking discussion (Figs. 2, 3). More porous matrices were formed when the concentration of urea was increased, which produced softer and more elastic aerogels. The Young's modulus of the aerogels varied between 6353 and 31,066 N/m^2, but the aerogel with the highest absorption capacity had a Young's modulus of 15,403 N/m^2. The obtained aerogel had better mechanical properties than lignocellulose aerogel prepared from lignocellulose/LiCl/DMSO (1250 N/m^2) (Liu et al. 2019). This result promises good physical properties comparable to that of other materials and excellent sorption performance, as discussed below.

The thermal behaviors of the coir fibers and cellulose aerogels were evaluated at 25–1000 °C under a nitrogen atmosphere. The results are shown in Fig. 8. It can be observed that for all samples, a small weight loss is detected between 25 and 100 °C due to the evaporation of water. The second weight loss in the temperature range of 250–330 °C for either coir fiber or cellulose aerogels represents the simultaneous decomposition of cellulose and lignin compounds. The third weight loss is a further decomposition of cellulose and lignin, which occurs in the temperature range of 330–780 °C for coir fiber, 330–750 °C for cellulose aerogel (1:0), and 330–675 °C for the other cellulose aerogels (1:3), (1:4), and (1:5). It is apparent that the temperature range for decomposition is higher when the lignin content is higher. The further weight loss curve then levels off slowly after the third weight loss stage, indicating that the aerogel was carbonized at high temperature. Coir fiber is carbonized at 780 °C with a char yield of 15%, while cellulose aerogel (1:0) is carbonized at 750 °C with a char yield of 9.3%, and cellulose aerogels (1:3), (1:4), and (1:5) is carbonized at 675 °C with an average char yield of 15%. The

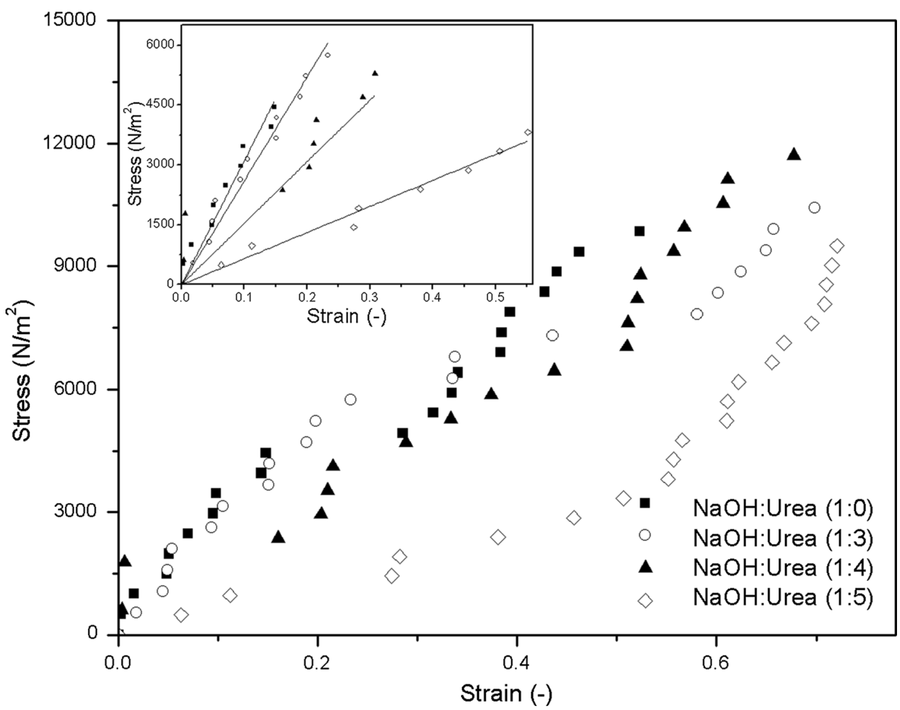


Fig. 7 Stress–strain curve of cellulose aerogels prepared from coir fibers

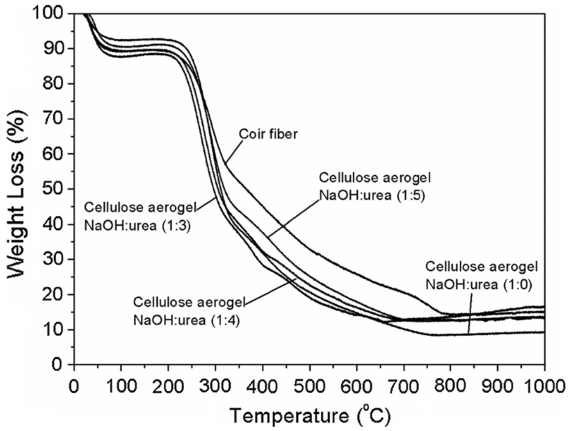


Fig. 8 Thermogravimetric analysis (TGA) results for coir fiber and cellulose aerogels

decomposition temperature of cellulose aerogel in this study is higher than that of materials prepared from coconut shell (200–400 °C) (Wan et al. 2015b), lignocellulose/LiCl/DMSO (200–315 °C) (Liu et al. 2019), and waste paper (230–330 °C) (Nguyen et al. 2014), which may be caused by the presence of lignin remaining in the pulp. Lignin has a complex structure and needs a higher temperature to decompose, and surprisingly, lignin improves the cellulose aerogel properties by having better thermal stability.

Water absorption

Figure 9 shows the effect of the NaOH:urea ratio on the absorption capacity of the aerogel. The absorption capacity of the aerogel increases proportionally to the urea concentration, reaching a maximum at a NaOH:urea ratio of 1:4 and subsequently decreasing. Figure 9 shows that at the first absorption, the aerogel was able to absorb up to 22 times its dry weight in water, which was the highest capacity. This value was higher than that of cellulose aerogel from waste paper (20 g/g) (Nguyen et al. 2014). At this point, the aerogel had the lowest density of approximately 0.0375 g/cm^3 and the highest porosity of 0.9755 (Fig. 6); hence, it was capable of absorbing much more water than the other aerogels. The results of the second and third absorption tests showed that the aerogel was still capable of absorbing a large volume of water. The highest absorption capacities of the second and third tests were also reached at a NaOH:urea ratio of 1:4, which was approximately 17 g/g. The absorption capacity was slightly reduced because the aerogel pore structure was damaged by squeezing. After the first test, the aerogel was squeezed to remove the absorbed water, and the squeezed aerogel was used for the second test. The procedure was repeated for the third test. After squeezing, the porous structure of the aerogel was substantially changed. However, the tests proved that the squeezed aerogel could absorb water again. Therefore, the aerogel is reusable and has good flexibility and durability.

Oil absorption

An oil absorption test was conducted for comparison with the water absorption test, and the results are

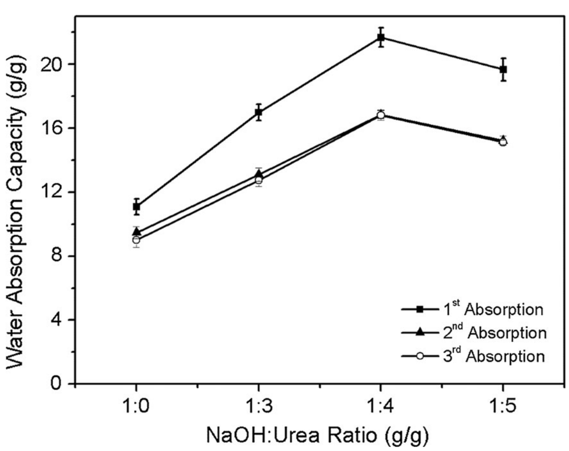


Fig. 9 Effect of the NaOH:urea ratio on the water absorption capacity of the cellulose aerogels

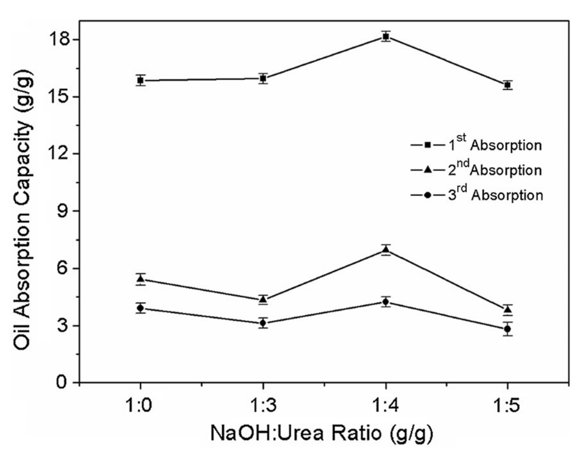


Fig. 10 Effect of the NaOH:urea ratio on the oil absorption capacity of the cellulose aerogels

shown in Fig. 10. The obtained results show a similar tendency to the results of the water absorption test. The maximum oil absorption capacity of the aerogel was reached at a NaOH:urea ratio of 1:4 and subsequently decreased, as observed in the water absorption profile. The highest oil absorption capacity was approximately 18 times the dry weight of the aerogel, which was lower than the water absorption capacity. This value was as high as the oil absorption capacity of cellulose aerogel from wheat straw (Li et al. 2014). Absorption depends on the capillary effect, van der Waals forces, pore morphology, and viscosity (Pääkkö et al. 2008; Gui et al. 2011; Wang et al. 2013). Lower viscosity greatly facilitated penetration into the porous network of the aerogel and resulted in a higher absorption capacity. The oil absorption capacity of the aerogel was lower than its water absorption capacity because the viscosity of oil is higher than that of water. In addition, cellulose has many hydroxyl (–OH) functional groups, which make the cellulose hydrophilic and lipophobic, which means cellulose tends to absorb water more than oil. Therefore, the oil absorption capacity was lower than the water absorption capacity. As shown in Fig. 10, the NaOH:urea ratio has no significant effect on the oil absorption capacity, which is approximately 16 g/g. This might be caused by the hydrophilic–lipophobic region of the cellulose aerogel, which inhibits the oil from diffusing into the internal pores of the cellulose fiber (Fig. 5c). In the case of water absorption, the hydrophilic region allows water to easily diffuse and penetrate the internal pores, resulting in a higher absorption capacity. Figure 10 also shows that the oil absorption capacities decreased during the second and third absorption tests and were approximately 7 g/g and 4 g/g, respectively. This can be accounted for by the collapse of the porous structure during squeezing. However, the aerogel was still capable of absorbing oil, which indicates its flexibility and durability.

Dye removal

Figure 11 shows the effect of monolith height on the breakthrough curve. MB was selected for the kinetic adsorption model approach because it is a kind of cationic dye that is widely used in the coloring industry. In general, as shown in the figure, the effluent concentration of MB starts at its lowest value in the early stage of the sorption process and increases with time. However, the breakthrough curves gradually shift to the right as the bed depth increases, which indicates a broader mass transfer zone at higher bed depths. The higher bed depth also provides more time for MB to make contact with the aerogel during adsorption, which results in a lower effluent concentration at the same operation time. It is apparent that the adsorption efficiency and performance are better at higher bed depths because of the increase in surface area, which provides more binding sites for adsorption.

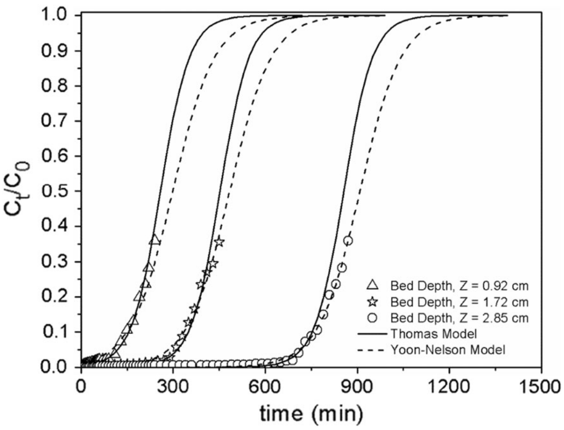


Fig. 11 Breakthrough curves of cellulose aerogels for methylene blue adsorption

For industrial applications, a continuous fixed-bed column is commonly used in the adsorption process instead of a batch system. Some kinetic adsorption parameters are needed for adsorption column design. Therefore, in this research, the experimental breakthrough curves were also fitted to the theoretical values to predict the concentration–time profile. Generally, there are two models, the Thomas and Yoon–Nelson models, which are used to obtain the kinetic adsorption parameters for a continuous fixed-bed column.

The Thomas kinetic model is commonly used to predict the maximum adsorption capacity of an adsorbent and the rate constant of adsorption (Thomas 1944). The expression of the Thomas adsorption model is given below:

$$\frac{C_t}{C_0} = \frac{1}{1 + \exp\left(k_{Th} \cdot q_e \cdot x/_{Q} - k_{Th} \cdot C_0 \cdot t\right)} \quad (5)$$

where k_{Th} is the rate constant of the Thomas model in mL mg^{-1} min^{-1}, q_e is the equilibrium MB uptake per g of adsorbent in mg g^{-1}, x is the total dry weight of the cellulose aerogel in g, and Q is the volumetric flow rate in mL min^{-1}. Two parameters (k_{Th} and q_e) are still unknown and are calculated by trial and error to solve the set equation (Eq. 4) in such a way that the sum of the square error is a minimum.

In the Thomas model, the maximum adsorption capacity is assumed to be equal to the equilibrium adsorption capacity (q_e). The fitted parameters of the Thomas adsorption model are tabulated in Table 1. The experimental data were well fitted to the Thomas model, given the small error and high value of R^2. Based on the Thomas model, the rate constant of MB adsorption to the cellulose aerogel (k_{Th}) was 0.00159 mL/mg min, and the maximum dye adsorption capacity was extraordinary, approximately 62,083.68 mg/g. This value is 60 and hundred times higher than the adsorption capacity of commercial activated carbon (980 mg/g) (Kannan and Sundaram 2001) and any adsorbent material synthesized from the other natural matters, such as phoenix tree leaf powder (146 mg/g) (Han et al. 2009), peanut husk (75.15 mg/g) (Song et al. 2011), and cellulose nanocrystal–alginate hydrogel (255.5 mg/g) (Mohammed et al. 2016), respectively.

The second model was introduced by Yoon–Nelson and expresses the breakthrough behavior by the mathematical formula shown below:

$$\frac{C_t}{C_0 - C_t} = \exp(k_{YN} \cdot t - \tau \cdot k_{YN}) \tag{6}$$

where k_{YN} is the rate constant of the Yoon–Nelson model in min^{-1} and τ is the time required for 50% adsorbate breakthrough in min (Han et al. 2009). Two parameters (k_{YN} and τ) are also unknown and are calculated by trial and error to minimize the square error for Eq. 5. The experimental points were also perfectly matched to the Yoon–Nelson model with a small error and high R^2, as tabulated in Table 1. The Yoon–Nelson model gives a rate constant (k_{YN}) of 0.0148 min^{-1} and different times to reach 50% of the breakthrough curves. The required time for 50% breakthrough increased as the bed depth increased from 299 to 484 and 909 min. This result shows that at the same operation time, the effluent concentration is lower with the higher bed depth. This phenomenon proves that higher removal efficiency can be achieved with higher bed depth.

The comparison of the experimental points and predicted curves made from the calculations based on both breakthrough models is shown in Fig. 11, confirming the square error and R^2 results in Table 1. Both of the predicted models matched the experimental points well. The lowest error value (*SS*) was given by the Thomas model, which means that the Thomas model was better at describing the process and behavior of MB adsorption into the cellulose aerogel. In addition, the Thomas model involves the dry weight of the aerogel used as the adsorbent and volumetric flowrate of the feed solution, which means it considers the volume and porosity of the aerogel. The porosity of an adsorbent is an important aspect that must be taken into account in calculating the kinetics of adsorption. The Yoon–Nelson model uses only the operation time, in addition to the influent and effluent concentration. Therefore, the Thomas model better described the adsorption behavior of the cellulose aerogel than the Yoon–Nelson model.

Table 1 Parameters of breakthrough curves for MB adsorption to cellulose aerogel

Z (cm)	k_{Th} (mL mg^{-1} min^{-1})	q_e (mg g^{-1})	R^2	*SS*
1. Thomas model				
0.92	0.00159	62,083.68	0.994	0.00076
1.72			0.993	0.00062
2.85			0.996	0.00023
Z (cm)	k_{YN} (min^{-1})	τ (min)	R^2	*SS*
2. Yoon–Nelson model				
0.92	0.0147	298.84	0.984	0.00125
1.72	0.0148	484.30	0.998	0.00033
2.85	0.0149	908.66	0.996	0.00019

Conclusion

A sulfur-free method to remove lignin from coir fibers is proposed in this research. The Kappa number of the fibers was significantly decreased when mechanical milling followed by sulfur-free alkali digestion at atmospheric pressure was used. The pulp resulting from this method was then successfully processed via a simple sodium hydroxide–urea solution method to produce a cellulose aerogel. NaOH–urea played an important role in the synthesis by transforming cellulose-I into cellulose-II structures and crosslinked the cellulose to produce the aerogel structure. The aerogel has a macroporous structure, ultralight density, high porosity, good durability, and thermal stability. The high sorption performance of the cellulose aerogel was proven by its high absorption capacities for water and oil, which were 22 and 18 times the dried weight of the aerogel, respectively. The aerogel also had a high adsorption capacity for MB of 62 g/g, which was one hundred times higher than that of any other adsorbents synthesized from another natural matters. Therefore, the obtained cellulose aerogel could be applied as an absorbent for any liquid spill, an adsorbent for dye removal, and thermal insulation.

Acknowledgments This work was supported by the Ministry of Research, Technology and Higher Education, Indonesia through a PMDSU Research Grant (Contract Numbers 15304/IT2.11/HK.00.02/2016, 77186/IT2.VII/HK.00.02/2017, and 798/PKS/ITS/2018). One of the authors (M.F.) would like to thank the Ministry of Research, Technology and Higher Education, Indonesia, for a doctoral scholarship through PMDSU. We thank Ms. Tiara Nur Pratiwi and Mr. Muhammad Abid Hidayatullah for their assistance with the experiments. We also thank Ms. Annie Mufyda Rahmatika for the TGA analysis.

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